A Project Report on

GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain

submitted in partial fulfillment of the requirements for the award of the degree of

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2025-2026

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CERTIFICATE

This is to certify that the project entitled "GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain" is a bonafide work of **Om Panchal (22106025), Sumant Bhise (22106095), Sidra Khan (22106028), Devang Dhuri (22106036)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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Project Report Approval for B.E.

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Abstract

GreenEquip is a decentralized, blockchain-enabled digital platform designed to enhance accessibility, transparency, and efficiency in the agricultural equipment ecosystem. The system facilitates seamless peer-to-peer interaction between equipment sellers and farmers through two primary functionalities: allowing sellers to list machinery for rent and enabling farmers to browse and book equipment based on their requirements. By simplifying the platform to these core operations, GreenEquip ensures ease of use while addressing key challenges such as lack of trust, inefficient resource utilization, and limited access to modern farming tools.

The platform incorporates a multi-portal architecture consisting of seller, farmer, and admin interfaces. Sellers can register, upload equipment details including category, region, pricing, age, and images, and manage their listings. Farmers can search for equipment, apply regional filters, and complete bookings by specifying rental duration and quantity. A structured booking workflow provides confirmation details and generates a downloadable receipt containing transaction information, including a blockchain-based transaction hash.

To ensure data integrity and trust, all transactions are recorded on a decentralized blockchain ledger, enabling transparent and tamper-proof record maintenance. Smart contract mechanisms facilitate secure and automated execution of bookings while preserving the decentralized nature of transaction processing. Although the system leverages decentralized blockchain infrastructure, all user-facing financial interactions are conducted in Indian Rupees (INR), with internal conversion mechanisms enabling compatibility with crypto-based transaction processing through MetaMask integration.

An administrative approval system is implemented to maintain platform quality and reliability. All equipment listings undergo verification and are categorized into pending, approved, or declined sections before becoming visible to farmers. Additionally, the admin dashboard provides analytical insights such as total equipment count, bookings, revenue, top-performing assets, and utilization metrics, ensuring effective system oversight alongside decentralized operations.

GreenEquip further integrates artificial intelligence components to enhance functionality. A Convolutional Neural Network (CNN) is used to validate uploaded equipment images and detect inconsistencies, triggering alerts for administrative review. Moreover, an intelligent chatbot assists farmers by analyzing soil and crop data to recommend suitable machinery, thereby supporting informed decision-making. The platform also supports regional language conversion, improving accessibility for diverse user groups.

Overall, GreenEquip aims to present a decentralized, secure, intelligent, and user-centric solution that leverages blockchain and AI technologies to modernize agricultural equipment access, improve trust among stakeholders, and promote efficient resource utilization.

Keywords: Decentralized Application (dApp), Smart Contracts, Ethereum, Blockchain, Peer-to-Peer Marketplace.

CONTENTS

1. Introduction 1
2. Literature Survey of Existing System 4
3. Limitation of Existing System 10
4. Problem Statement and Objective 13
   4.1 Problem Statement 13
   4.2 Objective 14
5. Proposed System 15
   5.1 Framework 15
      5.1.1 Seller Portal (Equipment Provider Interface) 16
      5.1.2 Farmer Portal (Equipment Consumer Interface) 17
      5.1.3 Admin Portal (System Control and Monitoring Interface) 19
      5.1.4 Backend Server (Application Core) 21
      5.1.5 Database Layer (MongoDB) 21
      5.1.6 Blockchain Layer (Decentralized Transaction System) 21
      5.1.7 AI Module (Intelligent System Component) 22
   5.2 Methodology 23
6. Experimental Setup 25
   6.1 Inputs 25
   6.2 Hardware and Software Setup 26
7. Results and Discussion 29
   7.1 Smart Contract Execution Results 29
   7.2 Blockchain Transaction Performance 30
   7.3 Portal-Based Functional Results 31
   7.4 Chatbot Recommendation System 32
   7.5 Evaluation Parameters 32
      7.5.1 Transaction Performance (Blockchain Evaluation) 32
      7.5.2 System Response Time 34
8. Conclusion and Future Work 36
   References 38
   Appendix 39
   Publication 42
   Acknowledgements 43

LIST OF FIGURES

Fig 5.1 : Architecture Diagram 15
Fig 5.2 : Listing New Equipment 17
Fig 5.3 : Farmer Portal Dashboard 18
Fig 5.4 : Equipment Booking on Farmer Portal 18
Fig 5.5 : Admin Dashboard 19
Fig 5.6 : Admin Dashboard Analytics 20
Fig 7.1 : Payment Receipt Generated 31
Fig 7.2 : Chatbot Assistant for Farmer & Language Translation 32

LIST OF TABLES

Table 2.1 : Literature review 7
Table 7.1 : Table representing relationship between gas fee and execution time 33
Table 7.2 : Average Time Taken for User Operations 35

LIST OF GRAPHS

Graph 7.1 : Graph showing the execution time of 5 tested transactions 33
Graph 7.2 : Graph representing gas fee variation across transactions.....34

ABBREVATION

dApp	*Decentralized Application*
ETH	*Ether*
DeFi	*Decentralized Finance*
DLT	*Distributed Ledger Technology*

Chapter 1

Introduction

1. Introduction

Agriculture remains a fundamental pillar of the global economy, particularly in developing countries such as India, where a significant portion of the population depends on farming for livelihood. Despite its importance, the agricultural sector continues to face numerous challenges, including limited access to modern machinery, high equipment ownership costs, and inefficient resource utilization. Small and marginal farmers, who constitute a large percentage of the farming community, are especially affected due to financial constraints and lack of technological support. As a result, there is a growing need for innovative solutions that can bridge the gap between resource availability and accessibility, while also ensuring trust and fairness in transactions.

One of the major issues in the agricultural domain is the underutilization of farm equipment. Agricultural machinery such as tractors, harvesters, and ploughing tools are often expensive to purchase and maintain. While some farmers or equipment owners possess such machinery, it remains idle for significant periods, leading to inefficient use of resources. At the same time, other farmers who require these tools are unable to access them in a timely and affordable manner. Traditional rental systems, where available, often suffer from lack of transparency, unreliable pricing, absence of proper verification mechanisms, and heavy dependence on intermediaries. These limitations reduce trust among stakeholders and hinder the effective adoption of shared agricultural resources.

With the advancement of digital technologies, there is an opportunity to transform the agricultural ecosystem by introducing platforms that enable efficient resource sharing. In particular, the integration of blockchain technology and artificial intelligence has opened new

avenues for building secure, transparent, and intelligent systems. A key advantage of blockchain technology lies in its decentralized nature, which eliminates the need for a single controlling authority and enables peer-to-peer interactions between users. Decentralization ensures that transaction data is distributed across the network, making it tamper-proof, verifiable, and resistant to manipulation. This is especially valuable in scenarios involving financial transactions and agreements between multiple parties who may not fully trust each other. On the other hand, artificial intelligence enables intelligent decision-making, automation, and data-driven insights, which can significantly enhance user experience and system efficiency.

In this context, GreenEquip is proposed as a decentralized, blockchain-based agricultural equipment platform that aims to simplify and modernize the process of accessing farm machinery. The platform is designed with a focused approach, supporting two primary functionalities: enabling equipment sellers to list their machinery for rent and allowing farmers to browse and book equipment according to their requirements. By leveraging decentralization at the transaction level, the platform ensures that all booking and payment records are securely maintained on a distributed ledger, thereby enhancing transparency and trust. At the same time, by limiting the system to these core operations, GreenEquip ensures simplicity, ease of use, and faster adoption among users with varying levels of digital literacy.

The platform adopts a multi-portal architecture consisting of seller, farmer, and administrative interfaces. Equipment owners can register on the platform and list their machinery by providing relevant details such as equipment type, region, pricing, age, and images. Farmers can search for equipment, apply filters based on location, and complete bookings through a structured workflow. This streamlined interaction reduces dependency on intermediaries and creates a direct connection between sellers and farmers, thereby improving efficiency and reducing costs while maintaining decentralized transaction handling.

To ensure trust and reliability, GreenEquip incorporates blockchain technology for recording all transactions. Each booking is securely stored on a decentralized ledger, generating a unique transaction hash that serves as proof of the transaction. This mechanism guarantees transparency, immutability, and accountability, which are fundamental characteristics of decentralized systems. Although the platform leverages decentralization for transaction management, it is designed to maintain user convenience by conducting all transactions in Indian Rupees. The system internally manages the conversion between fiat currency and blockchain-compatible processes, enabling users to benefit from decentralized security without requiring technical knowledge of cryptocurrencies. Integration with digital wallets such as MetaMask further enhances the security and authenticity of transactions.

An important aspect of the platform is its administrative control mechanism, which ensures quality and authenticity of listings. All equipment listings submitted by sellers are subject to approval by an administrator before becoming visible to farmers. This hybrid approach combines the benefits of decentralization with necessary centralized oversight, ensuring both trustless transaction execution and controlled data validation. Additionally, the admin interface provides analytical insights, including total equipment listings, number of bookings, revenue generated, top-performing equipment, and utilization statistics. These metrics enable effective monitoring and informed decision-making for platform management.

GreenEquip also integrates artificial intelligence components to enhance functionality and user experience. A Convolutional Neural Network (CNN) is employed to validate the images uploaded by sellers, ensuring that the images correspond to the selected equipment category. In cases of discrepancy, the system generates alerts for administrative review, thereby improving data accuracy. Furthermore, the platform includes an intelligent chatbot designed to assist farmers in selecting appropriate machinery. By analyzing soil characteristics and crop requirements, the chatbot provides personalized recommendations, helping farmers make informed decisions and optimize agricultural productivity.

Another key feature of the platform is its support for regional languages, which significantly improves accessibility for users from diverse linguistic backgrounds. By enabling the interface to be translated into languages such as Hindi, GreenEquip ensures that farmers with limited proficiency in English can effectively interact with the system. This inclusivity is crucial for promoting widespread adoption of digital solutions in rural areas.

In summary, GreenEquip aims to represent an innovative approach to addressing critical challenges in the agricultural sector through the integration of decentralized blockchain technology and artificial intelligence. By providing a secure, transparent, and user-friendly platform for equipment listing and booking, the system enhances resource utilization, builds trust among stakeholders, and empowers farmers with better access to modern tools. The proposed solution not only contributes to improving operational efficiency but also supports the broader goal of sustainable and technology-driven agricultural development.

Chapter 2

Literature Survey of Existing System

2. Literature Survey of Existing System

Smart Farming: Securing Farmers Using Block Chain Technology and IoT (Praveen Pappula, Mohammed Ali Shaik, Sampath Kumar, Tanupriya Choudhury, June 2021): This study examines the convergence of blockchain technology and the Internet of Things (IoT) as an integrated framework for reinforcing data security and operational transparency within agricultural systems. The work demonstrates that real-time farm data can be securely stored and systematically managed through distributed ledger infrastructure, thereby substantiating the principle of immutable record-keeping that underpins the present project. The findings provide direct empirical validation for the adoption of blockchain within GreenEquip as a mechanism for ensuring cryptographically secured and tamper-resistant rental transactions[1].

Blockchain-Based Smart Agriculture System (IRJMETS, 2023): This paper investigates the capacity of blockchain technology to enhance inter-party trust and transactional transparency within agricultural operational frameworks. It presents decentralized ledgers and smart contracts as foundational instruments for governing agricultural workflows and activity management. The mechanisms articulated therein are directly applicable to the present project, informing the design of equipment listing and rental agreement modules to ensure that all transactional records within GreenEquip remain fully verifiable and cryptographically secured[2].

Rental and Loan System in Agriculture Using Blockchain Technology (IJRASET, 2023): This study examines the applicability of blockchain technology to agricultural rental and

financial transaction systems, with particular emphasis on the utilization of smart contracts for the automated formalization of agreements and loan processing protocols. The work directly corroborates the foundational design rationale of the present project, wherein smart contract automation governs equipment rental workflows. GreenEquip extends the scope of this study by incorporating enhanced real-time asset tracking capabilities and a more interactive user interface[3].

An E-Commerce Platform of Agriculture Products Using Blockchain Technology (IJCRT, 2023): This paper investigates the deployment of blockchain-based platforms for facilitating secure digital transactions within the agricultural domain, with specific reference to the implementation of Ethereum smart contracts for payment and transaction management. The decentralized transactional framework demonstrated therein is extended within the present project and redirected toward peer-to-peer equipment rental operations, thereby ensuring secure, transparent, and disintermediated interactions between platform participants[4].

Blockchain Applications in Agriculture: Analytical Study (SSRN, 2024): This paper presents a comprehensive analytical examination of prevailing trends and emerging applications of blockchain technology across the agricultural sector, with particular emphasis on its capacity to improve traceability, transparency, and operational efficiency. The conceptual framework advanced in this study reinforces the theoretical underpinnings of GreenEquip's architectural design, affirming the criticality of decentralization as a foundational principle, notwithstanding the absence of a practical system implementation within the study itself[5].

Blockchain-Enabled Smart Agriculture System (MDPI Sensors, 2023): This study examines the integration of IoT sensor networks with blockchain infrastructure for the purposes of automated monitoring and secure data dissemination in smart farming environments. Although the present project does not incorporate IoT components, the methodological emphasis on secure and reliable data management is directly pertinent to the implementation of verifiable and tamper-resistant transactional integrity within the GreenEquip platform[6].

Consortium Blockchain-Based Agricultural Machinery Scheduling System (MDPI Sensors, 2020): This paper investigates the application of consortium blockchain architecture to the optimization of agricultural machinery scheduling, demonstrating how decentralized systems can enhance resource allocation efficiency. The resource management principles established in this work are operationalized within GreenEquip through real-time equipment availability querying and streamlined booking functionalities, with the present system further extending this scope to encompass integrated payment processing and equipment ownership tracking[7].

Blockchain Technology in Current Agricultural Systems: From Techniques to Applications (ResearchGate, 2020): This paper furnishes a comprehensive survey of blockchain applications within contemporary agricultural systems, delineating various distributed ledger techniques and their respective contributions to the improvement of transparency and inter-party trust. The study's findings align closely with the foundational objectives of GreenEquip, wherein blockchain infrastructure is systematically employed to ensure the security and verifiability of all rental transactions conducted on the platform[8].

Implementation of Smart Agricultural Equipment Rental System (IJASRET, 2022): This study examines the architectural design and operational workflows of conventional web-based agricultural equipment rental systems, providing substantive insights into user interaction paradigms and transactional process management. The identified absence of decentralization mechanisms within such centralized systems constitutes a significant architectural limitation, thereby providing empirical justification for the adoption of a blockchain-based approach in GreenEquip to achieve superior security, trust, and transactional integrity[9].

Metaheuristic and Machine Learning-Based Smart Engine for Renting and Sharing of Agriculture Equipment (ResearchGate, 2021): This paper investigates the application of metaheuristic optimization and machine learning methodologies to decision-making and resource utilization within agricultural equipment rental systems, with a focus on recommendation-driven efficiency improvements. Although the present project does not incorporate machine learning components, the overarching objective of operational efficiency enhancement is consonant with GreenEquip's approach of automating and streamlining rental workflows through smart contract execution and decentralized transaction management[10].

Table 2.1 : Literature Review

Year, Publication	Title	Domain	Algorithm	Research Gap
2021, Research Gate	Smart Farming: Securing Farmers Using Blockchain Technology and IoT	Smart Farming, IoT + Blockchain	Blockchain + IoT Integration, Smart Contracts	Focuses on security and monitoring; lacks rental, payment, and user interaction modules
2023, IRJMETS	Blockchain-Based Smart Agriculture System	Blockchain in Agriculture	Ethereum Smart Contracts, Decentralized Ledger	Limited real-world implementation; lacks scalability and real-time system evaluation
2023, IJRASET	Rental and Loan System in Agriculture Using Blockchain Technology	Agricultural Rental & Loan System	Blockchain + Smart Contracts	Does not include user-friendly interfaces or real-time tracking; limited system usability

2023, IJCRT	An E-Commerce Platform of Agriculture Products Using Blockchain Technology	Blockchain-based Agriculture E-commerce	Ethereum Blockchain, Smart Contracts	Focuses on product selling; lacks equipment rental and scheduling features
2024, SSRN	Blockchain Applications in Agriculture (Bibliometric/Analytical Study)	Blockchain in Agri-Food Systems	Bibliometric Analysis, Data Analytics	Theoretical analysis; lacks practical system implementation and prototype development
2023, MDPI Sensors	Blockchain-enabled Smart Agriculture System	IoT + Blockchain Agriculture	IoT Sensors + Blockchain Integration	High complexity; lacks evaluation in large-scale real-world agricultural environments
2020, MDPI Sensors	Consortium Blockchain-Based Agricultural Machinery Scheduling System	Machinery Scheduling	Consortium Blockchain + Scheduling Optimization	Focus limited to scheduling; lacks payment, rental ecosystem, and user interaction

2020, Research Gate	Blockchain Technology in Current Agricultural Systems: From Techniques to Applications	Blockchain in Agriculture	Comparative Analysis of Blockchain Techniques	Does not propose a working system; is a purely review-based study
2022, IJASRET	Implementation of Smart Agricultural Equipment Rental System	Rental System (Centralized)	Web-based System (Centralized Architecture)	No blockchain integration; lacks transparency, security, and trust mechanisms
2021, Research Gate	Metaheuristic and Machine Learning-Based Smart Engine for Renting and Sharing of Agriculture Equipment	ML-based Recommendation System	Metaheuristic Optimization + Machine Learning	Focuses only on recommendation ; lacks secure transactions, blockchain, and payment systems

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Existing agricultural equipment rental and management systems, as studied in the referenced literature [1]–[10], exhibit several limitations that hinder their effectiveness, scalability, and adoption among farmers, particularly in rural and semi-urban regions. These limitations arise due to technological constraints, lack of transparency, inefficient system design, and minimal integration of advanced computational techniques. A critical analysis of these systems reveals key gaps that justify the need for an improved solution such as GreenEquip.

One of the primary limitations observed in traditional and digital agricultural rental platforms is the lack of transparency in transactions. Many existing systems rely on centralized architectures where data is controlled by a single authority, leading to potential manipulation, data inconsistency, and lack of trust among users. Farmers often face uncertainty regarding pricing, availability, and transaction authenticity, as there is no immutable record of operations. As highlighted in [1], [3], and [6], centralized systems are prone to data tampering and do not provide verifiable transaction histories, thereby reducing user confidence.

Another major drawback is the dependence on intermediaries in conventional rental ecosystems. Middlemen often act as facilitators between equipment owners and farmers, leading to increased costs and inefficiencies. These intermediaries may introduce biased pricing, delays, and lack of accountability. Studies such as [2] and [5] emphasize that the involvement of third parties not only increases operational complexity but also reduces the direct benefits received by both equipment providers and farmers.

The underutilization of agricultural machinery is another persistent issue identified in existing systems. While some farmers own expensive equipment, it often remains idle for long

durations due to lack of proper platforms for sharing or renting. Conversely, farmers in need of such equipment are unable to access it efficiently. Existing platforms fail to provide a dynamic and accessible marketplace that ensures optimal utilization of resources. This limitation has been discussed in [4] and [7], where inefficient allocation mechanisms result in both economic loss and reduced agricultural productivity.

Additionally, many existing systems suffer from poor user interface design and lack of accessibility. A significant portion of the target user base, especially small and marginal farmers, may not be technologically proficient. Complex interfaces, lack of regional language support, and limited usability reduce adoption rates. As noted in [8], systems that do not consider user-centric design principles often fail to achieve widespread acceptance in rural communities.

Another critical limitation is the absence of intelligent decision-support systems. Most platforms operate as simple listing and booking systems without providing any form of recommendation or guidance to users. Farmers are required to manually select equipment without insights into suitability based on soil type, crop requirements, or environmental conditions. Research in [9] highlights the importance of integrating artificial intelligence to enhance decision-making, which is largely missing in existing implementations.

Furthermore, data validation and authenticity issues are prevalent in current systems. Equipment listings often lack proper verification, leading to inaccurate or misleading information. There is typically no mechanism to ensure that uploaded images or descriptions correspond to actual machinery. This creates trust issues and may result in fraudulent listings. As discussed in [3] and [10], the absence of automated validation mechanisms reduces the reliability of such platforms.

Scalability and performance also pose challenges. Many existing solutions are not designed to handle large-scale data or concurrent users efficiently. Centralized databases may become bottlenecks, leading to slower response times and degraded user experience. Additionally, systems that attempt to use blockchain often store excessive data on-chain, resulting in high transaction costs and reduced efficiency, as pointed out in [6] and [9].

Another limitation is the lack of secure and flexible payment mechanisms. Traditional systems rely on cash-based or loosely integrated digital payments, which may not provide transaction security or traceability. Even in some blockchain-based systems, the use of cryptocurrencies without user-friendly abstraction creates barriers for adoption, especially among users unfamiliar with such technologies. Studies such as [2] and [7] indicate that payment systems must balance security with usability to be effective.

Moreover, insufficient monitoring and analytics capabilities limit the effectiveness of existing platforms. Many systems do not provide administrators with actionable insights such as equipment utilization rates, revenue trends, or user activity patterns. This lack of data-driven decision-making hampers system optimization and long-term sustainability, as identified in [5] and [8].

Finally, most existing systems fail to adopt a hybrid architectural approach that combines the strengths of decentralized and centralized technologies. Purely centralized systems lack trust,

while purely decentralized systems often suffer from inefficiency and complexity. The absence of a balanced integration leads to suboptimal performance and limited practical applicability, as emphasized across multiple studies [1]–[10].

In summary, the limitations of existing systems include lack of transparency, reliance on intermediaries, inefficient resource utilization, poor accessibility, absence of intelligent features, data authenticity issues, scalability challenges, inadequate payment mechanisms, and limited analytics. These shortcomings highlight the need for a more robust, secure, and intelligent solution, thereby motivating the development of the GreenEquip platform, which aims to address these gaps through a hybrid, decentralized, and AI-integrated approach.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1 Problem Statement

Small and marginal farmers often struggle to access essential agricultural machinery due to high ownership costs and the inefficiencies of traditional rental systems, which lack transparency, proper verification, and intelligent support. At the same time, equipment owned by others remains underutilized, leading to resource wastage and reduced productivity. To address these challenges, the GreenEquip system proposes a blockchain-based platform that enables equipment sellers to list machinery for rent and allows farmers to easily search and book equipment. By integrating secure blockchain transactions, an admin-controlled approval system, INR-based payments with MetaMask support, and AI-driven features such as image validation and equipment recommendation, the platform ensures transparency, reliability, and efficient utilization of agricultural resources.

4.2 Objective

- **To Develop a Decentralized, Blockchain-Based Platform:** To design and implement a decentralized system that leverages blockchain technology to enable secure, transparent, and tamper-proof transactions between equipment sellers and farmers, while maintaining necessary administrative oversight for data validation and platform reliability.
- **To Enable Efficient Equipment Listing and Booking Mechanism:** To create a streamlined platform where sellers can list agricultural equipment with detailed attributes and farmers can easily search, filter, and book machinery based on region and requirements, thereby improving accessibility and utilization of resources.
- **To Automate and Secure Transactions Using Smart Contracts:** To implement smart contracts for automating booking and transaction processes, with MetaMask integration and INR-based payments supported through internal blockchain conversion.
- **To Integrate AI for Validation and Decision Support:** To incorporate a CNN model for equipment image validation and an intelligent chatbot that analyzes soil and crop data to recommend suitable machinery.
- **To Implement Admin Control with Analytics and Multi-Portal Access:** To develop seller, farmer, and admin portals with features such as listing approval, platform monitoring, and analytics including total equipment, bookings, revenue, and utilization metrics.

Chapter 5

Proposed System

5. Proposed System

5.1 Framework

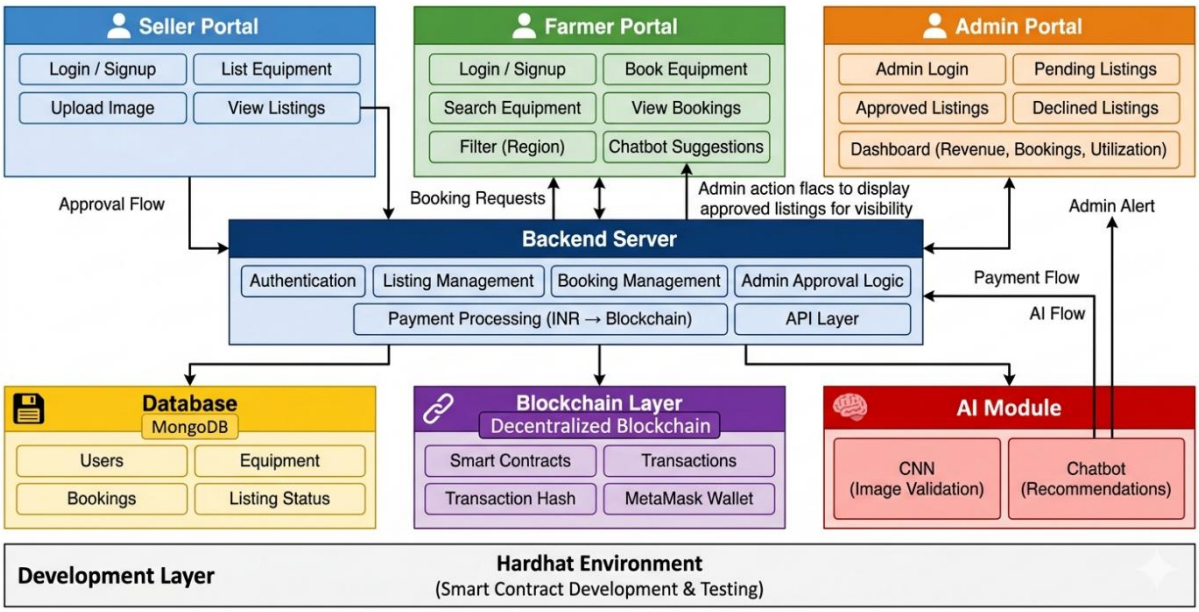


Fig 5.1: Architecture Diagram

The proposed system, GreenEquip, is a comprehensive, decentralized platform designed to modernize the agricultural equipment rental ecosystem by integrating blockchain technology, artificial intelligence, and a multi-portal web architecture. The system is structured around three primary user interfaces—Seller Portal, Farmer Portal, and Admin Portal—which are interconnected through a centralized backend, a NoSQL database, a decentralized blockchain layer, and intelligent AI modules.

The block diagram of the system illustrates the flow of data, control, and transactions between these components, highlighting how each module contributes to the overall functionality. The architecture ensures a seamless interaction between users while maintaining transparency, security, and efficiency. security.

5.1.1 Seller Portal (Equipment Provider Interface):

The Seller Portal is specifically designed for equipment owners who wish to list their agricultural machinery on the platform for rental purposes. It provides a structured and user-friendly interface that simplifies the process of onboarding and managing equipment listings.

Functional Components and Explanation:

1. Login / Signup:

   Sellers are required to register on the platform by providing necessary credentials. Once registered, they can securely log in to access their personalized dashboard. This authentication mechanism ensures that only authorized users can manage equipment listings.

2. List Equipment:

   Sellers can add new equipment to the platform by filling in detailed information such as equipment name, category, pricing, region, and condition. This structured input ensures uniformity and improves searchability for farmers.

3. Upload Image:

   Sellers must upload images of the equipment to provide visual verification. These images play a crucial role in building trust among users and improving transparency in listings.

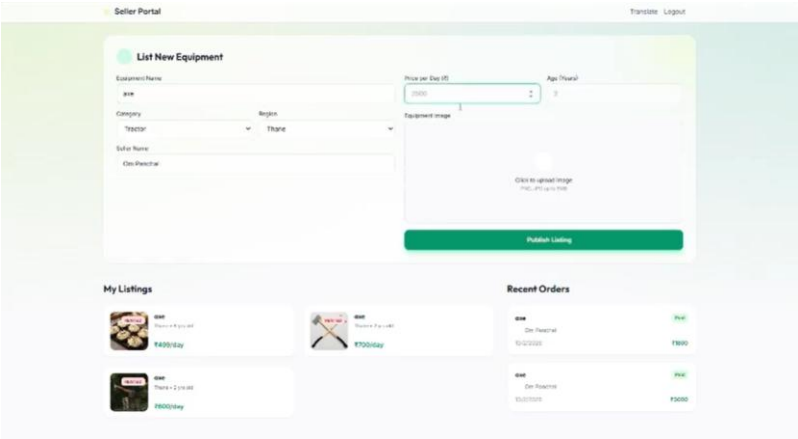


Fig 5.2 : Listing New Equipment

4. View Listings:

   Sellers can view all the equipment they have listed, along with their current status (pending, approved, or declined). This feature allows sellers to track the progress of their listings and make necessary updates if required.

Once a seller submits a listing, the data is transmitted to the Backend Server, where it undergoes processing. The uploaded image is passed through the CNN-based AI module for validation. After validation, the listing enters the Admin Approval Workflow, ensuring that only verified listings are published.

5.1.2 Farmer Portal (Equipment Consumer Interface):

The Farmer Portal serves as the primary interface for farmers to explore, evaluate, and rent agricultural equipment. It is designed to be intuitive and accessible, ensuring ease of use even for users with minimal technical expertise.

Functional Components and Explanation:

1. Login / Signup:

   Farmers must register and authenticate themselves to access platform services. This ensures secure transactions and personalized user experiences.

2. Search Equipment:

   Farmers can search for available equipment using keywords or categories. The search functionality is optimized to provide relevant results quickly and efficiently.

3. Filter by Region

Since agricultural equipment availability is location-dependent, farmers can filter listings based on their geographical region. This ensures that only nearby and accessible equipment is displayed.

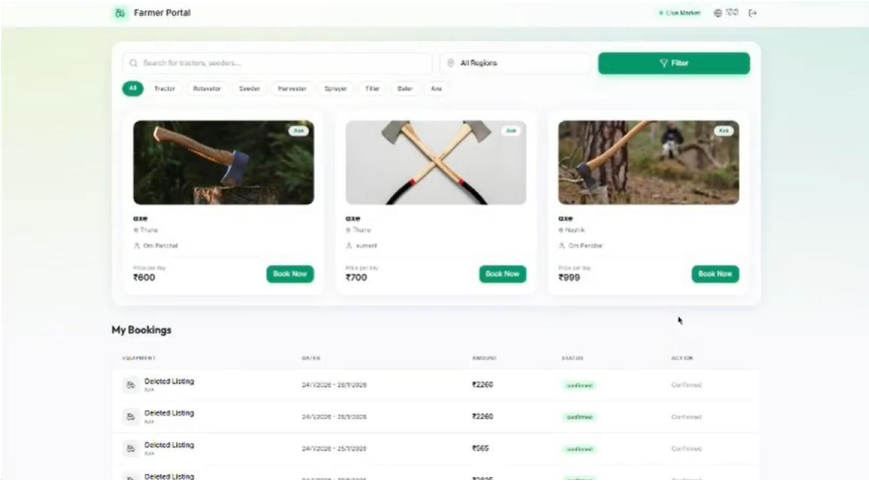


Fig 5.3 : Farmer Portal Dashboard

4. Book Equipment:

Farmers can select equipment and specify rental duration and quantity. The system calculates the total cost dynamically and presents it in INR, making it easy for users to understand pricing.

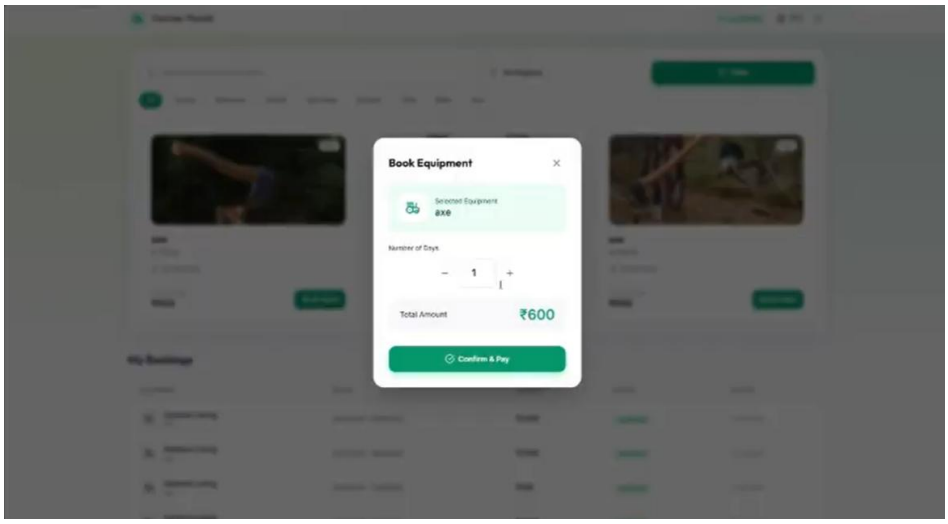


Fig 5.4 : Equipment Booking on Farmer Portal

5. View Bookings:

This feature allows farmers to access a complete history of their past and ongoing bookings. It includes details such as equipment name, rental duration, payment status, and transaction hash. This ensures transparency and provides a record for future reference.

6. Chatbot Suggestions:

The integrated AI chatbot assists farmers by recommending suitable equipment based on inputs such as crop type, soil conditions, and farming requirements. This reduces decision-making complexity and enhances productivity.

When a farmer initiates a booking:

- The request is sent to the Backend Server, where booking details are processed.
- The system interacts with the Database to fetch equipment data.
- Payment is processed through MetaMask, invoking the Blockchain Smart Contract.
- A transaction hash is generated and stored for verification.

5.1.3 Admin Portal (System Control and Monitoring Interface):

The Admin Portal plays a critical role in maintaining the integrity, reliability, and efficiency of the platform. It acts as a control center for managing listings and monitoring system performance.

Functional Components and Explanation:

1. Admin Login:

Only authorized administrators can access this portal through secure authentication, ensuring controlled access to system management features.

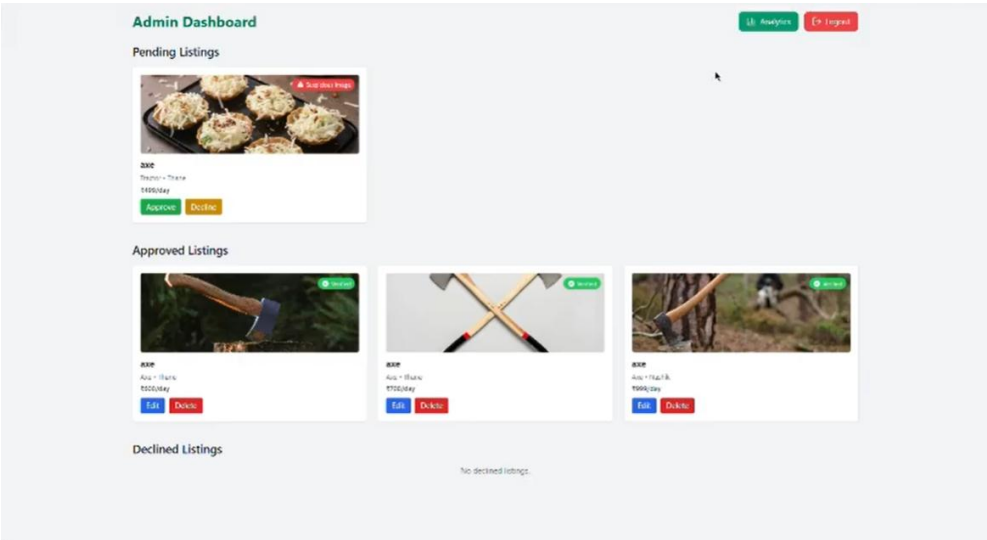


Fig 5.5 : Admin Dashboard

2. Pending Listings:

Newly submitted equipment listings appear in this section. The admin reviews these listings to verify their authenticity and completeness before approval.

3. Approved Listings:

Listings that meet all requirements are approved and made visible to farmers. This ensures that only verified and reliable equipment is available on the platform.

4. Declined Listings:

Listings that fail to meet quality standards or contain incorrect information are rejected. This helps maintain the credibility of the platform.

5. Dashboard Analytics:

The admin dashboard provides a comprehensive overview of system performance. It includes metrics such as total equipment listings, total bookings, revenue generated, and equipment utilization rates. These insights help in monitoring growth and making informed decisions.

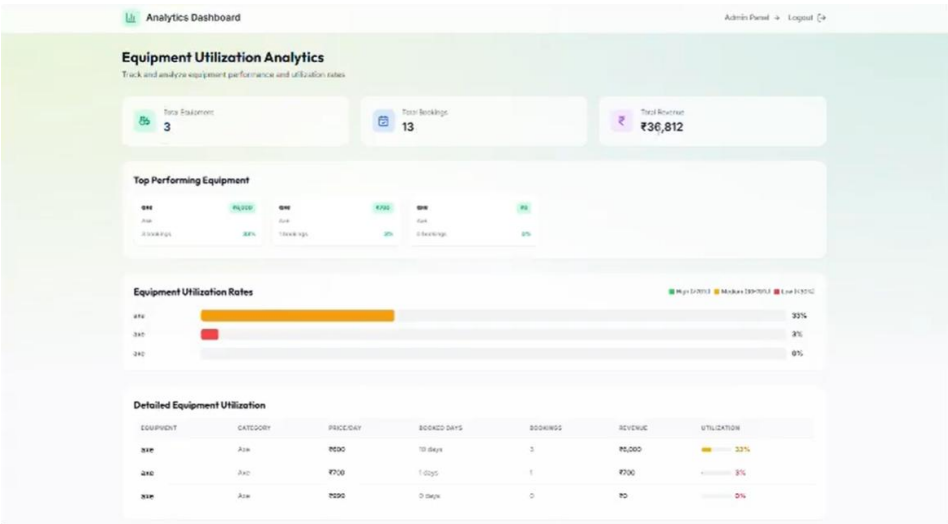


Fig 5.6 : Admin Dashboard Analytics

The Admin Portal receives:

- Listing data from the Backend Server
- Alerts from the AI Module (in case of image mismatch)

The Admin Portal sends:

- Approval or rejection decisions back to the backend are sent

- Updates that determine listing visibility in the Farmer Portal

5.1.4 Backend Server (Application Core):

The Backend Server is the central component that coordinates all system operations. It acts as an intermediary between the frontend portals, database, blockchain, and AI modules.

Functional Responsibilities:

- Authentication Management: Handles secure login and registration for all users.
- Listing Management: Processes equipment listings and forwards them for validation and approval.
- Booking Management: Manages booking requests, calculates costs, and updates booking records.
- Admin Approval Workflow: Routes listings to the admin and updates their status based on decisions.
- Payment Processing: Converts INR values into cryptocurrency equivalents and initiates blockchain transactions.
- API Communication: Ensures seamless communication between all system components.

5.1.5 Database Layer (MongoDB):

The MongoDB database is used to store and manage all system data efficiently. The stored data is:

- User Data: Includes information about farmers, sellers, and administrators, enabling personalized interactions.
- Equipment Listings: Stores detailed information about all listed machinery, including status and metadata.
- Booking Records: Maintains a history of all bookings, ensuring traceability and record-keeping.
- Listing Status: Tracks whether a listing is pending, approved, or declined, enabling controlled visibility.

5.1.6 Blockchain Layer (Decentralized Transaction System):

- The Blockchain Layer is responsible for handling secure and transparent financial transactions using the Ethereum network.
- Smart Contract: The smart contract governs transaction logic, ensuring that payments are executed only when predefined conditions are met.
- MetaMask Wallet Integration: Enables users to securely authorize and execute transactions.
- Transaction Hash: Each transaction generates a unique hash, which serves as immutable proof of the transaction and can be used for verification.

This layer ensures:

- Transparency in financial operations
- Security against data tampering
- Trust between buyers and sellers

5.1.7 AI Module (Intelligent System Component):

The AI Module enhances the functionality and intelligence of the platform through automation and predictive capabilities.

Components and Explanation:

- CNN-Based Image Validation:

  Analyzes uploaded equipment images to verify their authenticity and detect mismatches. This reduces fraudulent listings and improves data reliability.

- Chatbot Recommendation System:

  Interacts with farmers to understand their requirements and provides tailored equipment recommendations. This improves decision-making and user satisfaction.

The workflow begins with the equipment owner submitting equipment details to the GreenEquip application. These details include equipment specifications and the owner's wallet signature, which serves as cryptographic proof of ownership and intent. The application then forwards this data to the blockchain layer, where the equipment listing is securely recorded. Once the registration process is completed, confirmation messages and subsequent rental notifications are communicated back to the equipment owner through the application interface.

The Equipment Owner starts the process by entering information about the equipment into the GreenEquip program. The owner's wallet signature, which acts as cryptographic evidence of ownership and purpose, and equipment details are among these details. After obtaining this data, the GreenEquip application transmits the registration information to the blockchain layer, where the equipment listing is safely stored. The equipment owner receives confirmation messages and future rental notifications via the application interface after the registration process is finished.

The Equipment Renter uses the GreenEquip application to send search queries and rental requests in order to communicate with the system. The renter is shown listings of available equipment that are retrieved from blockchain data. The renter offers a wallet signature to approve the transaction when they choose an equipment item and move on with a rental request. The renter's unequivocal approval of the rental terms and related payment is ensured by this signature.

By gathering signed transaction data from both parties and sending it to the blockchain network, the GreenEquip application serves as an intermediary coordination layer. By authenticating payment data, monitoring equipment availability, and validating signatures, the blockchain verifies the transaction. The rental agreement is permanently entered into the distributed ledger after the transaction has been confirmed.

The GreenEquip application receives transaction status notifications from the blockchain following successful execution. These updates contain the equipment owner's payment confirmation and the renter's rental confirmation. In order to guarantee that both parties are promptly and consistently notified of the transaction outcome, the application then transmits this information to the corresponding users.

The technology removes the requirement for direct trust between equipment owners and renters by passing all crucial processes through the blockchain. Rental agreements and payments are safely enforced because to the unchangeable transaction logs and cryptographic wallet signatures. This interaction paradigm creates a dependable peer-to-peer leasing mechanism appropriate for decentralized agricultural equipment sharing, improves transparency, and lessens disagreements.

5.2 Methodology

The GreenEquip system adopts a decentralized, blockchain-driven architectural paradigm to systematically address the operational inefficiencies, transactional opacity, and inter-party trust deficiencies inherent in conventional agricultural equipment rental frameworks. The methodological design is structured to ensure secure, automated, and accessible interactions between smallholder farmers and equipment proprietors through a rigorously engineered digital platform.

The preliminary phase of the methodology entails a structured diagnostic analysis of the prevailing limitations associated with restricted access to affordable mechanization assets and the unreliability of extant rental mechanisms. Informed by this analysis, a decentralized solution is architected leveraging distributed ledger technology to systematically eliminate intermediary dependencies and establish verifiable trust through cryptographically secured, immutable record-keeping. The overall system architecture is partitioned into three primary functional layers: the Farmer Portal, the Seller Portal, and the Blockchain System, each fulfilling distinct operational roles within the platform's infrastructure.

The implementation phase commences with the development of a smart contract utilizing the Solidity programming language, which encapsulates and enforces the core business logic

governing platform operations. This smart contract algorithmically manages equipment asset registration, rental transaction execution, and programmable payment disbursement. Equipment proprietors authenticate and register their machinery by submitting relevant metadata, which is subsequently inscribed onto the blockchain ledger. Registered listings are thereafter accessible to farmers via a web-based interface, enabling asset selection in accordance with individual operational requirements and technical specifications.

Upon the initiation of a rental request by a prospective lessee, the system programmatically computes the aggregate rental cost as a function of the specified duration and renders this calculation for user confirmation. Following approval, the transaction is executed through a cryptographic digital wallet, specifically MetaMask, which interfaces directly with the deployed smart contract. The smart contract autonomously validates the transaction parameters, verifies the sufficiency of the submitted payment, and facilitates the secure disbursement of funds to the designated equipment proprietor. Concurrently, the complete transactional record is permanently and immutably inscribed onto the blockchain, ensuring full auditability and post-hoc verifiability.

The system maintains real-time data synchronization through continuous retrieval of updated state information from the blockchain network, enabling both farmers and equipment proprietors to monitor rental status, payment confirmations, and asset availability in real time. Furthermore, equipment imagery and ancillary metadata are administered through a dedicated storage layer, with corresponding references cryptographically linked to their respective blockchain records to preserve data coherence and integrity.

This methodological framework ensures that all platform operations are conducted in a transparent, cryptographically secured, and tamper-resistant manner, while concurrently enhancing operational efficiency through the comprehensive automation of rental workflows. Through the synergistic integration of distributed ledger technology with an ergonomically designed user interface, GreenEquip instantiates a practically viable and scalable solution for the modernization of agricultural equipment rental ecosystems.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Inputs

The GreenEquip system operates on real-time, user-generated inputs collected dynamically through the application interface. These inputs enable equipment listing, discovery, rental, and purchase operations within the platform.

Equipment owners provide essential details such as the name of the machinery, rental price, selling price, and their wallet address. Additional descriptive information and equipment images are also submitted through the interface and linked to the storage layer. Each equipment entry is assigned a unique identifier and recorded on the blockchain via the smart contract.

Farmers interact with the system by providing inputs such as selected equipment ID, transaction type (rent or buy), and rental duration (in case of renting). Based on these inputs, the system computes the total rental cost or retrieves the fixed selling price for purchase transactions. The farmer then confirms the transaction through their digital wallet.

The smart contract processes these inputs by validating the equipment details, verifying payment conditions, and executing the transaction. Upon successful validation, the payment is securely transferred to the equipment owner, and the equipment status is updated accordingly. All transaction records are stored on the blockchain, ensuring transparency and traceability.

The key input elements handled by the system include:

- Equipment details (name, rental price, selling price, availability)
- User wallet addresses (farmer and seller)
- Transaction parameters (rent or buy, duration if applicable)
- Transaction data (payment amount, timestamps) ecosystems

6.2 Hardware and Software Setup

Hardware Requirements: -

1. Client Device (End-User)

    - Any standard desktop, laptop, or mobile device capable of running a modern web browser.

    - No high-performance hardware is needed since the FarmChain dApp runs directly in the browser.

2. Internet Connection

    - A stable broadband or mobile internet connection is required.

    - Ensures smooth communication with the Ethereum blockchain for transaction processing and real-time data retrieval.

3. Development Machine (Developer-Side)

    - Processor: Dual-core CPU or higher.

    - Memory (RAM): Minimum 4 GB (8 GB recommended for optimal performance).

    - Storage: At least 10 GB free disk space for dependencies, local blockchain data, and project files.

    - Additional Tools: Modern code editor (e.g., Visual Studio Code), Node.js runtime, and Hardhat local blockchain environment.

Operating Environment:

1. Execution-Environment

The FarmChain application is designed to run primarily on the client-side, ensuring that users can interact with the platform without needing additional installations beyond a supported browser and the MetaMask extension. The following aspects define the execution environment:

- The frontend interface (developed using modern web technologies) executes entirely within the user's web browser.
- The chatbot logic and interaction layer also run client-side, leveraging browser-based JavaScript execution to deliver real-time responses and guidance to the user.
- Since computation and logic are handled locally on the client's browser, system requirements for end-users remain lightweight.

This client-side execution model not only improves performance and scalability but also reduces reliance on centralized servers, aligning with the decentralized nature of blockchain-based applications.

2. Blockchain-Network

The core functionality of the FarmChain dApp relies on Ethereum or other EVM-compatible blockchains. This ensures flexibility and portability across multiple blockchain ecosystems.

- Deployment Environment: In production, the application connects directly to the Ethereum mainnet or an EVM-compatible network to handle real-world transactions and smart contract execution.
- Development & Testing Environment: For developers, the system uses a Hardhat Network, a local Ethereum-like blockchain simulation. Hardhat provides:
  - Fast and reliable local blockchain testing.
  - Tools for compiling, deploying, and debugging smart contracts.
  - The ability to simulate real blockchain conditions (e.g., gas fees, block mining) without incurring actual costs.

Development Environment:

1. Smart Contract Framework

- Compiling Solidity smart contracts into deployable bytecode.
- Deploying contracts to local, testnet, or mainnet environments.
- Testing contract logic with built-in testing tools (using JavaScript/TypeScript).
- Debugging and error tracing, including stack traces, console.log outputs, and gas usage reporting.

- Simulating Ethereum nodes locally through the Hardhat Network, enabling developers to test features without incurring real gas costs.

This framework forms the backbone of the development workflow by ensuring reliable, repeatable, and cost-effective contract development.

2. Smart Contract Language

Solidity is the programming language used for developing the smart contracts that govern the FarmChain dApp. Key contracts include:

FarmChain.sol: The primary smart contract handling business logic and decentralized operations.

Lock.sol: A supporting contract, often used in scaffolding or managing time-based operations such as escrow or staking.

Solidity is specifically designed for the Ethereum Virtual Machine (EVM), making it the most suitable choice for ensuring secure and deterministic contract execution.

3. Frontend Library

- Ethers.js

  A lightweight yet powerful JavaScript library designed for blockchain interactions. It is responsible for:

  - Connecting the frontend with MetaMask or other Ethereum wallets.
  - Reading on-chain data from deployed contracts.
  - Sending transactions and monitoring blockchain events.
  - Providing abstraction over low-level JSON-RPC calls to simplify development.

4. Code Editor

- Integrated support for JavaScript, TypeScript, and Solidity development.
- Extensions for Hardhat, Ethers.js, and Solidity debugging.
- Built-in Git integration for version control and collaborative workflows

Chapter 7

Results and Discussion

7. Results and Discussion

7.1 Smart Contract Execution Results:

The deployment and testing of the FarmMachinery smart contract on a local blockchain environment (Hardhat) yielded consistent and reliable results. The contract successfully handled core functionalities such as machinery listing, rental transactions, and event logging.

Observed Functional Outcomes:

- Machinery Listing:

  The listMachinery() function allowed sellers to add machinery details to the blockchain. Each listing was assigned a unique ID using an incrementing counter, ensuring proper identification and traceability of equipment.

- Data Storage and Retrieval:

  The getMachineries() function successfully returned all stored machinery records, confirming that blockchain storage was functioning correctly and could be accessed when required.

- Rental and Sharing Transactions:

The rentOrShareMachinery() function executed transactions based on user selection (rent or share). The function validated payments using conditional checks:

- Ensured correct machinery ID
- Verified sufficient payment (rentPrice/sharePrice)

Funds were securely transferred to the equipment owner's wallet using Ethereum's transfer() mechanism.

- Event Emission and Logging:

The rent() function emitted a Rented event, capturing:

- Listing ID
- Farmer's address
- Rental duration
- Payment amount

This event was successfully logged and could be tracked via transaction receipts, providing transparency and auditability.

- Result Interpretation:

The smart contract demonstrated:

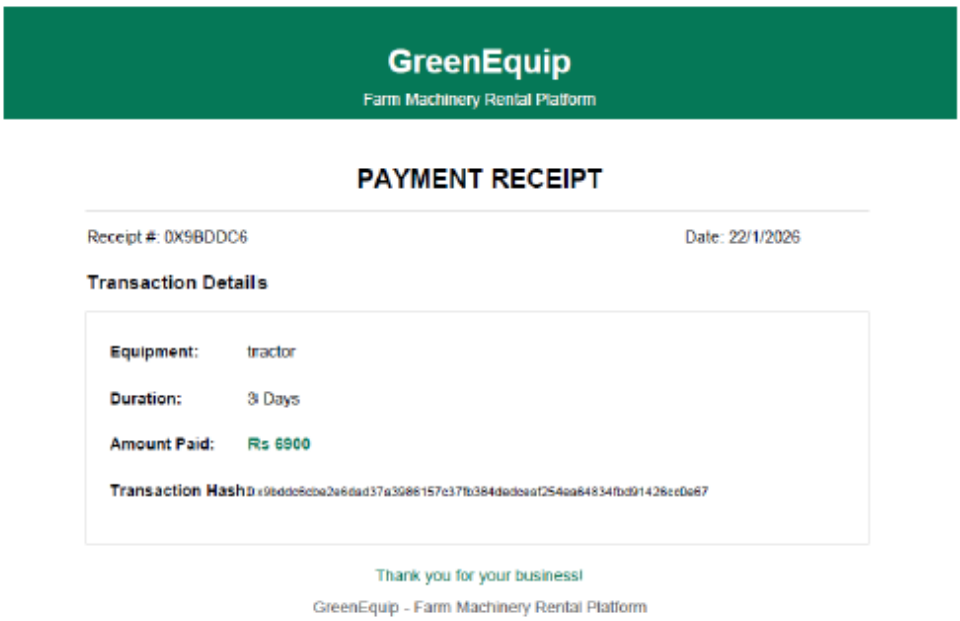
- Correct enforcement of payment conditions
- Secure transfer of funds without intermediaries
- Immutable storage of transaction-related data
- Reliable event generation for tracking activities

7.2 Blockchain Transaction Performance:

The integration of blockchain with the application through MetaMask was tested for multiple booking scenarios.

Observed Results:

- Transactions were successfully initiated from the Farmer Portal using MetaMask.
- Each transaction generated a unique transaction hash, which was:
  - Displayed to the user
  - Stored in the database
  - Included in downloadable receipts



GreenEquip

Farm Machinery Rental Platform

PAYMENT RECEIPT

Receipt #: 0X9BDDC6 Date: 22/1/2026

Transaction Details

Equipment: tractor

Duration: 3 Days

Amount Paid: Rs 6900

Transaction Hash: 0x9bdde8cba2a6dad37a3986157e37fb384dadcaaf254aa64834fbd91426cc0a67

Thank you for your business!

GreenEquip - Farm Machinery Rental Platform

Fig 7.1 : Payment Receipt Generated

- Transactions were immutable and verifiable on the blockchain network.
- Payment transfers were executed directly between farmer and seller wallets without third-party involvement.

7.3 Portal-Based Functional Results:

The Seller Portal was tested for listing creation and management. Its results are:

- Sellers were able to successfully register and log in.
- Equipment listings were created with complete details and images.
- Listings were correctly forwarded to the admin for approval.
- Sellers could view and track the status of their listings.

The Farmer Portal was evaluated for search, booking, and payment functionalities. Its results are:

- Farmers successfully searched and filtered equipment based on region.
- Booking functionality correctly calculated total cost based on duration.
- Payment through MetaMask was successfully executed.
- Booking details and transaction hashes were accurately recorded.
- Users could view booking history without inconsistencies.

The Admin Portal was tested for listing management and analytics. Its results are:

- Admins could view pending, approved, and declined listings.
- Approval and rejection actions were correctly reflected in the system.

- Only approved listings were visible to farmers.
- Dashboard metrics displayed accurate system statistics.

7.4 Chatbot Recommendation System:

The Chatbot was implemented. Its results are:

- Farmers received equipment suggestions based on input conditions.
- Recommendations were relevant to crop and soil requirements.

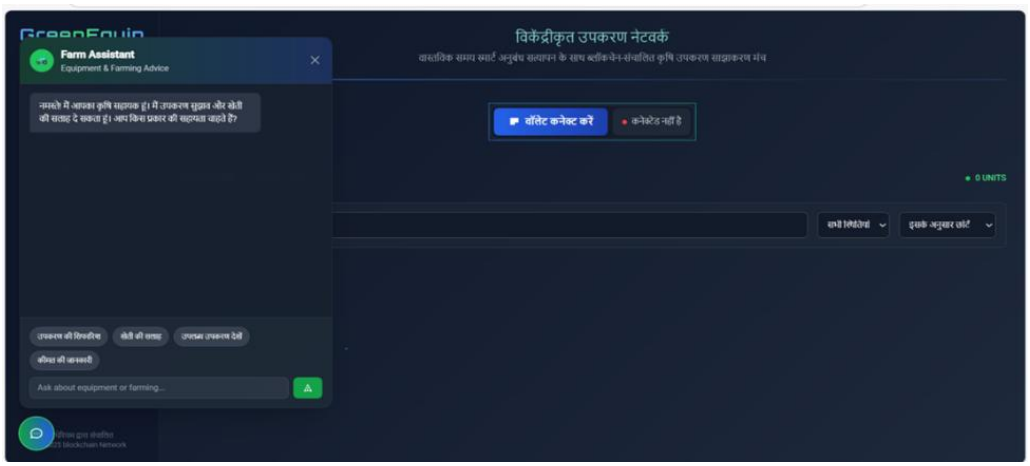


Fig 7.2 : Chatbot Assistant for Farmer & Language Translation

7.5 Evaluation Parameters

7.5.1 Transaction Performance (Blockchain Evaluation):

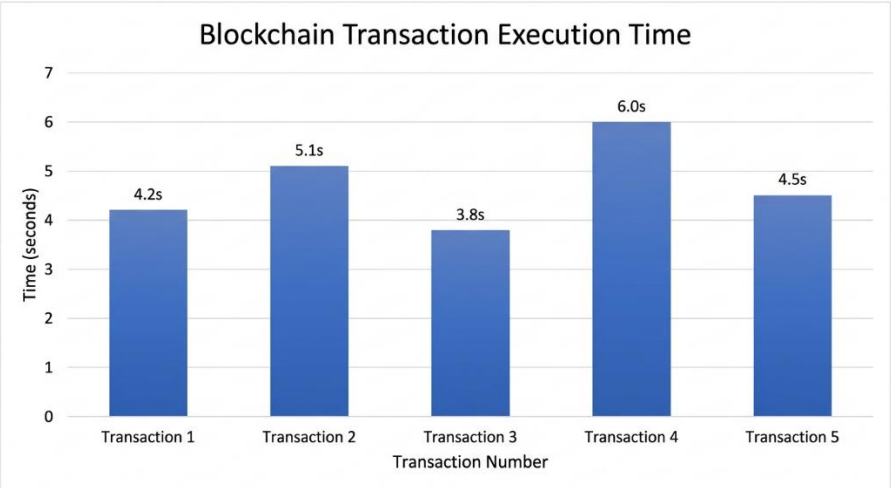
This evaluates how efficiently your system handles payments through blockchain.

Parameters:

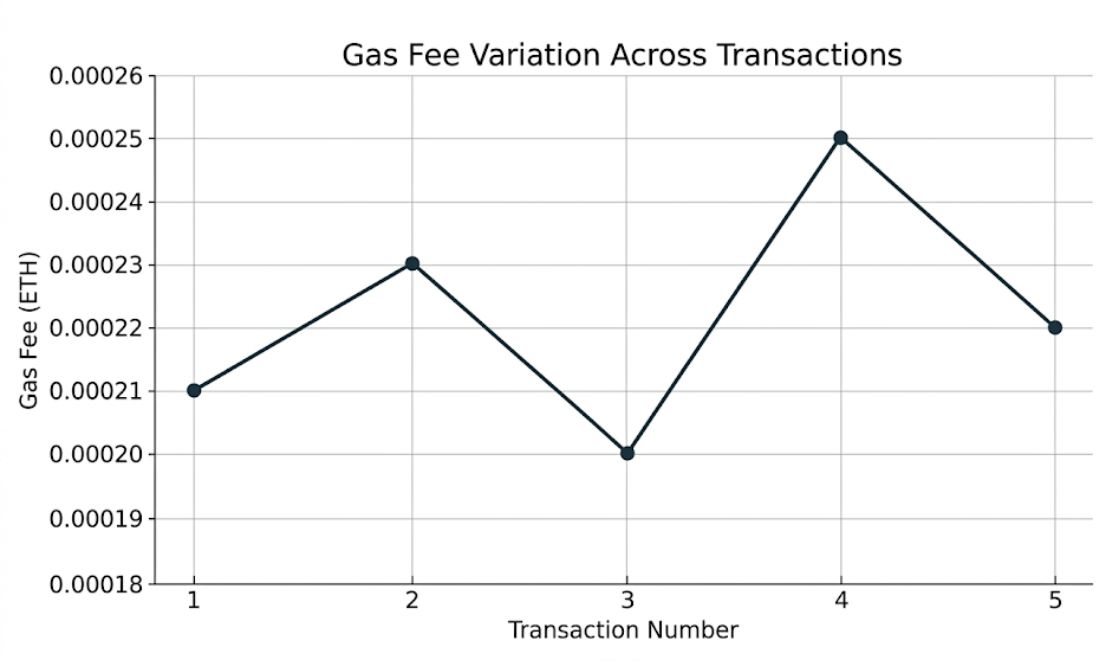
- Transaction Execution Time (seconds):
  Time taken from MetaMask confirmation → transaction mined.
- Gas Fee (ETH / INR):
  Cost required to execute smart contract functions.
- Transaction Success Rate (%):
  Number of successful transactions vs total attempts.

Table 7.1 : Table representing relationship between execution time and gas fee

Transaction No.	Execution Time (s)	Gas Fee (ETH)	Status
1	4.2	0.00021	Success
2	5.1	0.00023	Success
3	3.8	0.00020	Success
4	6.0	0.00025	Success
5	4.5	0.00022	Success



Graph 7.1 : Graph showing the execution time of 5 tested transactions



Graph 7.2 : Graph representing gas fee variation across transactions

Derived Metrics:

- Average Execution Time: ~4.72 seconds
- Success Rate: 100%
- Average Gas Fee: ~0.00022 ETH

7.5.2 System Response Time:

Parameters:

- Login Response Time
- Equipment Search Time
- Booking Processing Time

Table 7.2 : Average time taken for user operations

Operation	**Avg Time (ms)**
Login Authentication	320 ms
Equipment Search	450 ms
Booking Request	620 ms
Listing Submission	700 ms

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

The GreenEquip system effectively illustrates how blockchain technology may be used to construct a decentralized agricultural equipment rental business. Through smart contracts, the developed system facilitates safe equipment listing, rental execution, and payment settlement, guaranteeing that all transactions are publicly documented and automatically enforced without human interference. The findings demonstrate that rental activities can be carried out dependably while preserving transaction consistency and data integrity.

Through the implemented three-layer architecture, the system effectively separates user interaction from transaction enforcement. Farmers were able to browse equipment, calculate rental costs, and complete bookings with immediate confirmation, while equipment owners could list assets and receive payments directly upon successful rental execution. Blockchain-based transaction handling ensured that equipment availability, rental status updates, and payment transfers were synchronized across both user portals in real time.

The experimental process verified that using immutable ledger records prevented frequent problems like double booking and payment conflicts and decreased ambiguity in rental agreements. Every rental that was finished produced a verified transaction record that gave both parties accountability and traceability. These results demonstrate the system's capacity to function as a peer-to-peer, trustless marketplace.

Overall, the findings confirm that implementing blockchain-powered rental systems in agricultural settings is feasible. GreenEquip provides a scalable solution for increasing access

to vital farming resources by demonstrating how decentralized automation may improve efficiency, transparency, and dependability in equipment sharing.

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Appendix

The following Solidity smart contract implements the core functionality of the GreenEquip system, including equipment listing and transaction handling (rental and purchase):

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

contract FarmMachinery {

    struct Machinery {

        uint256 id;

        string name;

        uint256 rentPrice;

        uint256 sellPrice;

        address owner;

    }

    Machinery[] public machineries;

    uint256 public machineryCounter;

    function listMachinery(string memory _name, uint256 _rentPrice, uint256 _sellPrice) public {

        machineries.push(Machinery(machineryCounter, _name, _rentPrice, _sellPrice, msg.sender));

        machineryCounter++;

    }
```

```
function transactMachinery(uint256 _machineryId, bool _isRent) public payable {
        require(_machineryId < machineries.length, "Invalid machinery ID");
        Machinery storage machinery = machineries[_machineryId];
        if (_isRent) {
            require(msg.value >= machinery.rentPrice, "Insufficient funds for rent");
        } else {
            require(msg.value >= machinery.sellPrice, "Insufficient funds for purchase");
        }
        payable(machinery.owner).transfer(msg.value);
    }
    function getMachineries() public view returns (Machinery[] memory) {
        return machineries;
    }
}
```

The system uses MongoDB to store application data. Below is a representation of the collections used.

1. User Collection:

```
{
  "user_id": "U101",
  "name": "Ramesh Patil",
  "email": "ramesh@example.com",
  "role": "farmer",
  "region": "Maharashtra"
}
```

2. Equipment Collection:

```
{
  "equipment_id": "E201",
  "name": "Tractor",
  "category": "Heavy Machinery",
  "price_per_day": 1500,
  "owner_id": "U102",
  "status": "approved"
```

}

3. Booking Collection:

{
"booking_id": "B301",
"equipment_id": "E201",
"farmer_id": "U101",
"days": 3,
"total_cost": 4500,
"transaction_hash": "0xABC123..."

Publication

The research work associated with this project has been submitted for publication in the IEEE International Conference on Emerging Trends in Information, Communication & Systems (ICETICS-2026), organized by the Bansal Institute of Science and Technology, Bhopal. The conference is technically sponsored by the IEEE Madhya Pradesh Section and serves as a prominent platform for researchers, academicians, and industry professionals to present innovative solutions in emerging technologies.

The submitted paper is currently under the review process and is expected to be presented at the conference scheduled to be held on 10–11 July 2026 in hybrid mode. Upon acceptance and presentation, the paper will be considered for inclusion in the IEEE conference proceedings and potential indexing in IEEE Xplore, subject to meeting quality and formatting requirements.

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